

Normative Hip Range of Motion and Strength Profiles of Uninjured Baseball Athletes: A Systematic Review

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Context: The baseball athlete incorporates the use of the entire kinetic chain while throwing. Hip range of motion (ROM) and strength have been previously studied in its relation to injury development. It is a clinical necessity to understand the normative profiles of hip musculoskeletal function across the playing career. **Objectives:** To identify the normative hip ROM and strength profiles of baseball players across level of play. **Evidence Acquisition:** The researchers searched the electronic databases MEDLINE, SPORTDiscus, CINAHL, and Embase using a custom search strategy. After applying inclusion and exclusion criteria, 31 articles were reviewed. Of these 31 articles, 22 examined hip ROM and/or strength in uninjured baseball athletes. A manual search of the references of those 22 articles revealed 4 additional articles that met the inclusion criteria. In total, 26 articles were included in data collection. **Evidence Synthesis:** Articles were assessed for quality using the appraisal tool for cross-sectional studies. The appropriate hip ROM and strength data, along with player demographics, were extracted and reviewed to ensure accuracy. Articles were also categorized based on level of play and player position. **Conclusions:** Hip ROM and strength profiles of baseball athletes across all levels of play were identified and presented to comprehensively summarize the available data on normative musculoskeletal hip profiles. Observed differences between positions and across levels of play were documented. In general, hip external rotation ROM was greater than internal rotation across all levels and between the stance and lead limbs. As age and level of play increased, the difference between hip external rotation and internal rotation ROM also increased. Hip strength increased as level of play (and subsequently age) increased. These summarized data can be used as benchmarks to compare clinical populations.

Keywords: musculoskeletal, overhead athlete, throwing athlete, mobility, lower extremity

Key Points

There are observed differences between hip range of motion (ROM) and strength profiles of baseball athletes when considering limbs, position, and level of play. Methodology varies greatly from study to study, garnering direct statistical comparisons difficult. It is a clinical necessity to understand the normative musculoskeletal profiles of baseball athletes to aid in the screening of and subsequent plan of care development.

Context

Baseball is a popular sport in the world: the United States alone reported roughly 16.6 million participants in 2023, and globally, 75 countries have organized men's World Baseball Teams as of 2024.^{1,2} Common injuries in baseball can be divided into either noncontact or contact injuries. Noncontact injuries encompass multiple joints across the body, and commonly reported contact injuries include sport-related concussion, ankle sprains, and fractures.³ Injury surveillance at the youth level demonstrates both contact and noncontact mechanisms with a high prevalence of upper quarter injuries.^{3,4} For both high school and college level, noncontact injuries to both the shoulder and elbow account for 38%

to 58% of all reported injuries, primarily due to overuse.⁴⁻⁶ Likewise, professional baseball athletes experience shoulder and elbow injury rates ranging from 51% to 58% of injuries.^{4,7} The observed rate of injuries across all levels of baseball requires further inquiries into how the entire kinetic chain particularly the lower extremities contribute to time-loss injuries incurred from baseball participation.

The throwing motion utilizes the entire kinetic chain, involving a combination of coordinated movements from the upper and lower extremities as well as the trunk to produce large amounts of force and torque during the throwing motion.⁸ The shoulder and elbow are subjected to supraphysiologic kinetics, initiating at the late cocking phase, with the shoulder reaching up to 165° of external rotation (ER) and forces exceeding 300 N at the shoulder and elbow.⁹ The kinetic chain theory postulates that segmental forces generated in a proximal-to-distal fashion can effect subsequent joints where a "catch-up" phenomenon is described to articulate the load-sharing principles of the body during movement.^{10,11} Of this series of links in the kinetic chain, the hip plays a

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pivotal role in the throwing motion. During the cocking phase, knee and hip-flexion shifts the center of gravity in a posterior direction. In the follow-through phase, as the throwing arm decelerates, the transfer of forces is transmitted to the stride foot, requiring high demand on the hip posterior musculature to decelerate the body.¹² Additionally, the overhead throwing athlete generates large, multi-planar ground reaction forces equal to body weight at push-off and up to one and a half times body weight at landing.¹³ These observed ground reaction forces have been shown to correlate with upper-extremity kinetics and kinematics, as well as ball velocity in baseball athletes.^{13,14} These cumulative findings demonstrate the importance of the lower-extremity during the overhead throwing motion on performance-related factors and the potential for injury development.

It is crucial to assess how both lower extremity strength and range of motion (ROM) can affect the overhead throwing athlete. This hip joint has gained particular interest in the examination of throwing athletes for both performance and injury-related concerns. Decreased strength at the hip has been linked to throwing-related pathomechanics, suggesting that weakness can lead to inefficient force transfer with resultant increase in upper-extremity loading.^{15,16} Likewise, the loss of hip ROM has demonstrated increased odds of developing time-loss injuries.^{17–21} Prior to understanding the variations in musculoskeletal hip profiles that may affect performance or injury development, it is imperative to understand normative ROM and strength values across all playing levels. In doing so, sports medicine professionals can screen for deviations from previously published normative ROM and strength values. While previous studies have demonstrated differences in ranges of hip strength and motion between stance and lead limb across various levels of play, the methods employed and outcomes in uninjured baseball players vary widely, making clinical comparison difficult.^{17,20,22–27}

It is a clinical necessity to access the data related to hip musculoskeletal function for all levels of baseball to help inform clinical decision making. Therefore, the purpose of this systematic review is to describe the normative data in uninjured baseball athletes by position across all levels of play. Specifically, we focused on the variables of hip ROM and strength between the stance and lead limbs.

Methods

Design

Prospective registration was sought through PROSPERO. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were followed during the creation of this review.

Evidence Acquisition

An online literature search was performed by the lead author within MEDLINE, SPORTDiscus, Embase, and CINAHL Complete on September 30, 2023. The search was updated on April 18, 2024 to ensure no new articles could be missed. A population, intervention, and outcome model was instituted to help guide the search strategy. Two authors developed the search strategy, which involved keywords and subject headings (Medical Subject Headings and Emtree functionality) for each database, with the goal of retrieving the best set of possible results. This strategy helps to capture all results relevant to the research question. The “population” included all levels of baseball players; the “intervention” included specific measurement devices use to collect the main outcome measures

of the study, and the “outcome” included the musculoskeletal variables of interest. Each of the search terms under population, intervention, and outcome were first combined using “OR.” Consequently, a set of results were generated for each part of the population, intervention, and outcome model. The authors then used “AND” to bring all the population, intervention, and outcome results together (Table 1). Lastly, manual searches were conducted using the reference lists of the identified articles.

For a study to be included in this review, it needed to identify hip ROM and strength profiles among healthy baseball players. The study designs associated with these types of investigations are level 3 evidence according to the Oxford Center for Evidence-Based Medicine 2011 Levels of Evidence. Baseball players could be defined as professional, college, or adolescent. This review allowed for a variety of different competition levels so the hip profiles between the different levels of play could be documented. Articles had to meet the following inclusion criteria: (1) investigated an uninjured baseball cohort, (2) the abstract was available in a literature search, (3) hip ROM measured with goniometry or inclinometry, and/or strength outcomes measured with handheld dynamometry or isokinetic testing, (4) full article access in the English language, and (5) included pitchers and/or positions players. Studies were excluded if they investigated the effect of an intervention on ROM or strength (levels 1 and 2 evidence), included injured baseball players, did not report means for the variables of interest, were published in conference proceedings, and published prior to 2003. Articles published prior to 2003 were excluded as the game of baseball, including style of play, optimal training protocols, and rules, have changed drastically over the past 2 decades.²⁸

Following article selection, the quality of findings and potential bias were assessed using the appraisal tool for cross-sectional studies (AXIS). The AXIS tool was developed to appraise cross-sectional studies. The AXIS has 20 questions accompanied by explanatory help text that provides background knowledge on what the question is asking. Two authors (Hannon and Morrow) independently rated each article that met the specified inclusion criteria. A third author (Kennedy) was introduced to the appraisal process when any disagreements between the authors could not be resolved (19% of articles needed a third review).

Six authors collaborated to extract the appropriate data. The extracted data included player demographics, position, hip ROM and strength measures for both lead and stance limbs, devices used to measure the outcomes, patient position, unit of measure, and any reported reliability metrics. The means and SDs were extracted from all studies. Data were organized into tables by the primary author to enhance visualization of included strength and ROM data. Hip ROM data were presented in individual tables by level of play, while hip strength was presented in 1 table due to the small number of included studies. All hip ROM and strength data were synthesized by limb (stance and lead) to better ensure understanding of interlimb differences. Additionally, the data were organized by playing position. Using the data from the included studies, heterogeneous populations and data collection methods prevented the authors from conducting a meaningful meta-analysis.

Evidence Synthesis

Following the completion of the literature search, a total of 26 articles met the specified inclusion criteria and were categorized as level 3 evidence according to the Oxford Center for Evidence-Based Medicine 2011 Levels of Evidence (Figure 1). All articles were cross-sectional in nature, investigating a cohort of baseball players

Table 1 Search Strategy for Each of the Databases Included Within the Review Process

Step	Strategy	MEDLINE	SPORTDiscus	CINAHL	Embase
S44	S13 AND S20 AND S43	121	101	111	144
S43	S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42	91,772	26,739	42,012	96,061
S42	Isokinetic hip strength	10	9	7	12
S41	Isokinetic strength	1409	1256	833	1775
S40	Isotonic hip strength	0	0	0	0
S39	Isotonic strength	69	52	31	100
S38	Isometric hip strength	51	42	43	49
S37	Isometric strength	2452	1621	1194	2919
S36	Hip abduction strength	167	124	136	208
S35	Hip adduction strength	41	33	28	47
S34	Hip internal rotation strength	0	1	0	3
S33	Hip external rotation strength	37	34	30	46
S32	Hip-flexion strength	74	29	48	119
S31	Hip extension strength	65	50	58	87
S30	Hip strength	509	382	370	680
S29	Hip rotation range motion	39	31	35	39
S28	Hip internal rotation range motion	20	17	17	20
S27	Hip external rotation range of motion	5	6	7	5
S26	Hip adduction range of motion	10	8	5	10
S25	Hip abduction range of motion	27	10	13	30
S24	Hip extension range of motion	27	19	19	27
S23	Hip-flexion range of motion	65	58	56	74
S22	Hip range of motion	826	354	483	961
S21	Range of motion	88,159	24,074	39,981	91,284
S20	S14 OR S15 OR S16 OR S17 OR S18 OR S19	22,081	14,430	12,617	37,704
S19	Isokinetic dynamomet*	2604	2071	1497	3460
S18	Isokinetic	9447	9005	5416	12,345
S17	Inclinomet*	1189	465	655	1546
S16	Goniomet*	4661	1635	4179	9066
S15	Handheld dynamom*	773	361	485	1072
S14	Dynamometer	10,033	6033	4171	19,021
S13	S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12	5132	75,807	3418	6062
S12	Adolescent baseball	143	101	107	149
S11	Little league baseball	35	474	24	45
S10	Minor league baseball	2423	1437	1636	88
S9	Major league baseball	605	15,738	314	584
S8	Professional baseball	580	3612	376	670
S7	High school baseball	206	559	135	226
S6	Youth baseball	202	305	136	208
S5	College* baseball	327	2756	229	326
S4	Baseball pitcher*	945	1672	640	1066
S3	Baseball athlete*	98	103	67	102
S2	Baseball player*	1452	23,347	880	2020
S1	Baseball	5132	75,807	3418	6059

that included an uninjured group. AXIS scores ranged from 12 to 15, with 20 being the highest possible score. The critical appraisal of all 26 articles can be found in Table 2. Higher total scores represented better quality studies. Answers that met the criteria for each question

were labeled as “yes,” and answers that did not meet the criteria were labeled as “no” or “not applicable.” It is important to note that questions 7, 13, and 19 of the AXIS relate to nonresponders and are not applicable to the single time point, observational studies

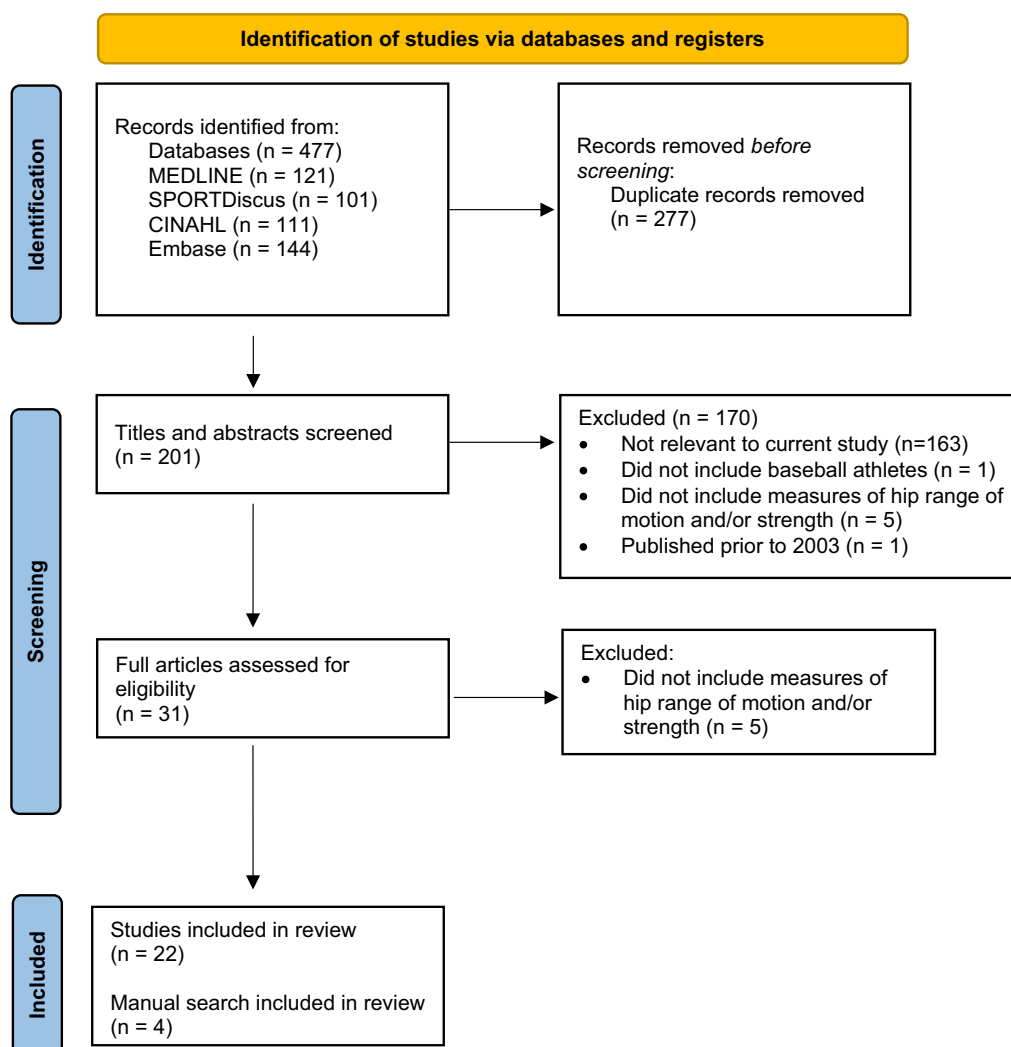


Figure 1 — Flow diagram for article selection. PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only. Adapted from Page et al.²⁹

included in this review. Overall, none of the included studies in this review had significant conflicts of interests related to the outcomes of each study. Only a total of 5 articles had an adequate sample size justification in the form of a formal power analysis.^{31,39,40,42,43}

This systematic review presents hip ROM and strength data on a total of 3947 uninjured baseball players. One study included up to 141 youth female baseball athletes but did not specify the inclusion of females in a “Normal Group” versus an “Injury Group;” and therefore, the total number of females included in the final review could not be determined.³⁶ Of the 3947 total athletes in this review, 9 studies reported on youth athletes (2766 total; 173 pitchers and 141 non-pitchers players).^{25,30,32,36–38,42–44} Four youth baseball athlete studies did not differentiate between position group,^{36,38,42,43} and 1 study only differentiated youth and high school pitchers with unknown allotment for nonpitchers by level.³⁰ Four studies reported on high school athletes (133 total; 105 pitchers and 28 nonpitchers).^{25,30,31,39} Eight studies reported on college athletes (169 total; 151 pitchers and 18 nonpitchers).^{23,24,33,35,45–48} Seven studies included professional baseball players (840 total; 446 pitchers and 394 nonpitchers).^{17,22,26,34,40,41,49} Detailed study characteristics are presented in Table 3.

All studies investigating ROM (n = 26) were reported using degrees (Table 4). The methodology employed varied from study

to study, with differences in instrumentation, number of raters, and rater position reported. Hip internal rotation (IR) ROM was reported in all 26 studies; hip ER ROM was reported in 25 studies, and hip total rotational ROM was reported in 20 studies. All hip ER and IR ROM was measured in either a supine or seated position with both the hip and knee bent to 90° of flexion or a prone position with the knee flexed to 90°. Hip abduction and adduction ROM were measured in 2 studies.^{41,46} Robb et al⁴¹ found significant side to side differences in hip abduction (lead hip, 35.7° [10.7°], vs stance hip, 43.4° [12.0°]) and hip adduction (lead hip, 31.6° [6.2°], vs stance hip, 50.8° [8.4°]). Comparatively, no significant side to side differences were found in a study investigating hip abduction (lead hip, 48.0° [6.3°] vs stance hip, 48.7° [6.6°]) and adduction (lead hip, 18.0° [5.5°] vs stance hip, 19.4° [4.8°]) in college baseball pitchers.⁴⁶ Hip-flexion ROM was reported in 2 studies.^{42,46} In a group of 89 uninjured youth baseball athletes, Saito et al⁴² reported similar hip-flexion ROM in the lead hip (122.0° [12.4°]) and the stance hip (121.0° [12.3°]). Likewise, Yanagisawa et al⁴⁶ found no significant differences between lead hip flexion (139.2° [6.8°]) and stance hip flexion (138.7° [6.2°]) in a cohort of 23 uninjured college baseball pitchers. Hip extension ROM was measured in 1 study and reported no significant differences in

Table 2 Quality of Evidence Assessed Using AXIS for Cross-Sectional Studies

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Total
Becket et al ³⁰																					14
Bullock et al ³¹																					14
Camp et al ¹⁷																					13
Camp et al ²²																					13
Cheatham et al ²³																					13
Crawford et al ²⁴																					12
Harding et al ³²																					14
Ishigaki et al ³³																					14
Laudner et al ³⁴																					14
Laudner et al ³⁵																					14
Nagasawa et al ³⁶																					13
Oliver and Weimar ³⁷																					14
Oliver et al ³⁸																					14
Picha et al ²⁵																					14
Plummer et al ³⁹																					15
Plummer et al ⁴⁰																					15
Robb et al ⁴¹																					14
Saito et al ⁴²																					15
Sakata et al ⁴³																					15
Sauers et al ²⁶																					14
Sekiguchi et al ⁴⁴																					14
Shimamura et al ⁴⁵																					14
Yanagisawa et al ⁴⁶																					14
Zeppieri et al ⁴⁷																					14
Zeppieri et al ⁴⁸																					14
Zipser et al ⁴⁹																					14

Note. Green, Yes; Yellow, Not Applicable; Red, No.

ROM between limbs in college baseball pitchers (lead hip, 22.4° [3.2°], vs stance hip, 23.4° [5.0°]).⁴⁶ Finally, a passive straight leg raise was measured in the supine position in 2 studies involving youth baseball athletes.^{42,44} Minimal differences in passive straight leg raise angles were observed between limbs in both studies, including one study with 89 uninjured participants (lead hip, 74.8° [14.8°], vs stance hip, 75.1° [14.1°])⁴² and another with 161 in a “without pain” group (lead hip, 66.0° [8.0°] vs stance hip, 68.0° [8.0°]).⁴⁴

A total of 10 studies assessed isometric hip strength measures in competitive baseball players,^{25,30,32,34,38,40,45–47,49} with 4 articles reporting data solely on pitchers.^{32,45–47} Study populations varied

across these studies, with 4 studies examining hip strength in players under the age of 18 years,^{25,30,32,38} 3 studies assessing college players,^{45–47} and 3 studying professional players.^{34,40,49} Study methods varied considerably with hip abduction strength being the most consistently assessed in 8 of the 10 studies.^{25,30,32,34,45–47,49} Hip internal and ER were also reported in 5 of the studies.^{25,32,38,45,46} Hip abduction, ER, and IR strength values can be found in Table 5. Isometric hip extension strength was measured via handheld dynamometry in college baseball.^{45,46,49} Both Yanagisawa et al⁴⁶ (stance hip: 590.9 [156.9] N, lead hip: 572.6 [139.7] N) and Zeppieri et al⁴⁷ (stance hip: 77.4 [11.5] N·m, lead hip: 74.6 [10.2] N·m) found

Table 3 Study Characteristics for All Included Articles

Study	Level of play	Position group	N	Pitchers	Nonpitchers	Age, y	Height, cm	Weight, kg	Clinical measures	Unit of measure	Reliability reported
Beckett et al ³⁰	Youth and high school	Pitchers and nonpitchers	108	Youth: 38 High school: 34	36	Youth: 9.9 (1.5) High school: 15.1 (1.3)	Youth: 145 (13.4) High school: 177.2 (17.4)	Youth: 38.9 (10.9) High school: 70.4 (12.7)	Passive ER and IR ROM with standard goniometer; isometric hip abduction strength with HHD	ROM: Deg strength: N/kg	No
Bullock et al ³¹	High school	Pitchers	32	32	0	16.3 (1.2)	184.0 (6.9)	76.8 (20.8)	Passive ER and IR ROM with bubble level goniometer	ROM: Deg	No
Camp et al ¹⁷	Professional	Pitchers and nonpitchers	258	129	129	NR	NR	NR	Passive ER and IR ROM with standard goniometer	ROM: Deg	No
Camp et al ²²	Professional	Pitchers and nonpitchers	96	54	42	Pitchers: 27.9 (4.7) Nonpitchers: 27.3 (3.6)	Pitchers: 188.4 (5.0) Nonpitchers: 186.3 (4.6)	Pitchers: 97.4 (11.1) Nonpitchers: 96.3 (7.7)	Passive ER and IR ROM with standard goniometer	ROM: Deg	No
Cheatham et al ²³	College	Pitchers	29	29	0	20 (1.4)	190 (6.0)	89.3 (10.7)	Passive ER and IR ROM with inclinometer	ROM: Deg	Yes
Crawford et al ²⁴	College	Pitchers	11	11	0	20.4 (1.6)	191.4 (6.0)	92.9 (8.0)	Passive ER and IR ROM with standard goniometer	ROM: Deg	No
Harding et al ³²	Youth	Pitchers and nonpitchers	28	28	0	13.9 (2.9)	167.9 (20.0)	61.3 (18.6)	Passive ER and IR ROM with standard goniometer; isometric hip ER, IR, and abduction strength with HHD	ROM: Deg strength: kg	Yes
Ishigaki et al ³³	College	Pitchers and nonpitchers	22	4	18	20.4 (1.0)	173.5 (5.8)	72.4 (8.8)	Passive ER and IR ROM with inclinometer	ROM: Deg	Yes
Laudner et al ³⁴	Professional	Pitchers and nonpitchers	80	40	40	Pitchers: 23.5 (2.9) Nonpitchers: 23.9 (2.1)	Pitchers: 190 (4.6) Nonpitchers: 184.4 (6.3)	Pitchers: 90.3 (8.1) Nonpitchers: 88.9 (10.6)	Passive ER and IR ROM with inclinometer; isometric hip abduction strength with HHD	ROM: Deg strength: kg	No
Laudner et al ³⁵	College	Pitchers	34	34	0	20 (1.3)	187.2 (5.8)	86.9 (6.8)	Passive ER and IR ROM with inclinometer	ROM: Deg	Yes
Nagasawa et al ³⁶	Youth	Pitchers and nonpitchers	2133	NR	NR	NR	140.8 (9.8)	36 (10.7)	Passive ER and IR ROM with standard goniometer	ROM: Deg	No
Oliver and Weimar ³⁷	Youth	Pitchers	26	26	0	11 (1)	152 (9)	48 (11)	Passive ER and IR ROM with inclinometer	ROM: Deg	Yes
Oliver et al ³⁸	Youth	NR	22	NR	NR	13.0 (2.0)	165.8 (14.5)	59.7 (14.7)	Passive ER and IR ROM with digital inclinometer; isometric hip ER and IR strength HHD	ROM: Deg strength: N	Yes
Picha et al ²⁵	Youth and high school	Pitchers and nonpitchers	72	Youth: 8 High school: 14	Youth: 19 High school: 28	Youth: 10 (1) High school: 15 (12)	Youth: 140 (13) High school: 176 (13)	Youth: 37 (11) High school: 69 (20)	Passive ER and IR ROM with standard goniometer	ROM: Deg	Yes

(continued)

Table 3 (continued)

Study	Level of play	Position group	N	Pitchers	Nonpitchers	Age, y	Height, cm	Weight, kg	Clinical measures	Unit of measure	Reliability reported
Plummer et al ³⁹	High school	Pitchers	25	25	0	16 (1)	180 (6)	75 (9)	Passive ER and IR ROM with inclinometer	ROM: Deg	Yes
Plummer et al ⁴⁰	Professional	Pitchers and nonpitchers	170	86	84	Pitchers: 22.0 (2.3) Nonpitchers: 21.2 (2.0)	Pitchers: 188.3 (6.3) Nonpitchers: 181.6 (5.6)	Pitchers: 91.8 (11.5) Nonpitchers: 85.1 (10.3)	Isometric hip abduction strength with HHD	Strength: asymmetry ratio, lead leg stronger than stance leg	Yes
Robb et al ⁴¹	Professional	Pitchers	19	19	0	NR	192 (1)	93 (11)	Passive ER and IR ROM with bubble level goniometer	ROM: Deg	Yes
Saito et al ⁴²	Youth	Pitchers and nonpitchers	89	NR	NR	11.9 (2.0)	NR	NR	Passive ER and IR ROM with standard goniometer	ROM: Deg	Yes
Sakata et al ⁴³	Youth	Pitchers and nonpitchers	242	NR	NR	9.2 (1.3)	NR	NR	Passive ER and IR ROM with standard goniometer	ROM: Deg	No
Sauers et al ²⁶	Professional	Pitchers and nonpitchers	99	50	49	22 (2.8)	187 (5.4)	81.6 (7.7)	Passive ER and IR ROM with standard goniometer	ROM: Deg	No
Sekiguchi et al ⁴⁴	Youth	Pitchers and nonpitchers	161	39	122	11.0 (0.8)	NR	NR	Passive ER and IR ROM with standard goniometer	ROM: Deg	No
Shimamura et al ⁴⁵	College	Pitchers	29	29	0	20 (1.4)	188 (6) cm	89.3 (10.7)	Passive ER and IR ROM with inclinometer; isometric hip ER, IR, extension, and abduction strength with HHD	ROM: Deg strength: kg	Yes
Yanagisawa et al ⁴⁶	College	Pitchers	23	23	0	19.3 (1.0)	177.4 (4.0)	72.9 (3.9)	Passive ER and IR ROM with standard goniometer; isometric hip flexion, extension, abduction, adduction, and ER and IR strength with HHD	ROM: Deg strength: N	Yes
Zeppieri et al ⁴⁷	College	Pitchers	14	14	0	19.4 (1.4)	189.0 (6.4)	96.1 (1.4)	Passive ER and IR ROM with bubble level goniometer; isometric hip abduction and extension strength with HHD	ROM: Deg strength: N·m	Yes
Zeppieri et al ⁴⁸	College	Pitchers	7	7	0	19.57 (1.1)	1.89 (0.6)	93.3 (6.8)	Passive ER and IR ROM with bubble level goniometer	ROM: Deg	No
Zipser et al ⁴⁹	Professional	Pitchers and nonpitchers	118	68	50	21.6 (2.0)	184.4 (6.8)	90.4 (13.3)	Isometric hip abduction strength with HHD	Strength: N·m	Yes

Abbreviations: Deg, degrees; ER, external rotation; HHD, handheld dynamometer; IR, internal rotation; ROM, range of motion.

Table 4 Hip Rotational Range of Motion Across All Levels of Play

Study	Position group	Lead ER	Lead IR	Lead TROM	Stance ER	Stance IR	Stance TROM	Position
Youth								
Nagasawa et al ³⁶	Pitchers and nonpitchers	47.0	49.0	NR	47.0	49.0	NR	Supine
Saito et al ⁴²	Pitchers and nonpitchers	NR	46.9 (1.2)	NR	NR	46.9 (13.3)	NR	Supine
Sakata et al ⁴³	Pitchers and nonpitchers	61.1 (7.6)	38.2 (12.2)	NR	60.4 (6.1)	40.1 (23.8)	NR	Supine
Sekiguchi et al ⁴⁴	Pitchers and nonpitchers	54.0 (11.0)	44.0 (13.0)	NR	49.0 (10.0)	48.0 (13.0)	NR	Supine
Oliver and Weimar ³⁷	Pitchers	37.2 (6.6)	28.5 (6.1)	65.7 (8.4)	35.0 (6.2)	31.3 (8.3)	66.4 (10.6)	Seated
Oliver et al ³⁸	NR	40.3 (7.3)	34.2 (9.5)	74.6 (11.6)	41.6 (5.8)	36.8 (9.2)	78.5 (12.2)	Seated
Beckett et al ³⁰	Pitchers and nonpitchers	38.7	38.4	77.1	37.7	40.8	78.5	Prone
Harding et al ³²	Pitchers	30.3 (7.3)	33.1 (8.9)	63.4 (8.8)	30.0 (7.5)	30.2 (6.6)	59.8 (9.6)	Prone
Picha et al ²⁵	Pitchers	26.7 (8.9)	42.6 (6.4)	69.3 (9.2)	30.4 (7.1)	37.9 (4.1)	68.3 (9.8)	Prone
	Nonpitchers	30.1 (4.8)	39.1 (7.5)	69.2 (9.3)	30.8 (5.7)	35.9 (9.1)	66.7 (11.8)	Prone
Saito et al ⁴²	Pitchers and nonpitchers	NR	49.1 (12.5)	NR	NR	49.4 (12.6)	NR	Prone
High school								
Bullock et al ³¹	Pitchers and nonpitchers	43.8 (11.4)	29.3 (7.9)	73.0 (15.2)	43.8 (11.0)	29.8 (8.1)	73.5 (14.0)	Supine
Plummer et al ³⁹	Pitchers	32.9 (6.9)	24.2 (6.8)	57.0 (11.0)	33.1 (6.6)	25.1 (8.2)	58.2 (13.4)	Seated
Beckett et al ³⁰	Pitchers and nonpitchers	38.9	34.4	73.2	38.5	33.1	71.6	Prone
Picha et al ²⁵	Pitchers	31.3 (5.7)	32.3 (9.2)	63.5 (9.2)	30.7 (7.7)	30.2 (5.5)	60.9 (9.3)	Prone
	Nonpitchers	29.2 (6.4)	31.6 (8.5)	60.7 (9.3)	28.9 (7.0)	30.9 (6.9)	57.7 (11.5)	Prone
College								
Crawford et al ²⁴	Pitchers	32.0 (8.0)	33.0 (7.0)	66.0 (1.0)	32.0 (5.0)	31.0 (6.0)	63.0 (9.0)	Supine
Ishigaki et al ³³	Pitchers and nonpitchers	29.9 (6.2)	34.4 (8.0)	NR	33.5 (6.1)	33.9 (6.6)	NR	Supine
Cheatham et al ²³	Pitchers	42.3 (11.9)	33.8 (7.8)	76.1 (9.8)	40.0 (11.7)	33.3 (9.4)	73.3 (10.5)	Seated
Shimamura et al ⁴⁵	Pitchers (right-handed)	39.4 (10.3)	35.6 (8.1)	NR	36.9 (9.8)	33.6 (9.4)	NR	Seated
	Pitchers (left-handed)	45.2 (13.6)	32.1 (7.4)	NR	43.2 (13.6)	33.0 (9.5)	NR	Seated
Ishigaki et al ³³	Pitchers and nonpitchers	42.0 (5.8)	34.0 (9.2)	NR	43.7 (6.5)	35.1 (8.4)	NR	Prone
Laudner et al ³⁵	Pitchers	35.8 (3.4)	27.7 (5.9)	63.4 (7.0)	35.4 (6.2)	28.9 (7.2)	64.2 (8.4)	Prone
Shimamura et al ⁴⁵	Pitchers (right-handed)	46.3 (12.1)	27.0 (8.9)	NR	32.2 (8.0)	24.8 (8.6)	NR	Prone
	Pitchers (left-handed)	48.9 (13.6)	17.3 (7.9)	NR	49.4 (7.3)	29.5 (8.4)	NR	Prone
Yanagisawa et al ⁴⁶	Pitchers	61.0 (7.5)	41.8 (7.2)	102.8 (8.9)	56.7 (6.5)	45.7 (5.8)	102.3 (9.1)	Prone
Zeppieri et al ⁴⁸	Pitchers	29.4 (8.9)	16.9 (8.5)	46.3 (11.5)	34.3 (7.5)	12.6 (3.6)	46.9 (5.6)	Prone
Zeppieri et al ⁴⁷	Pitchers	33.4 (9.8)	17.1 (15.1)	50.6 (14.8)	37.1 (7.0)	15.1 (4.6)	50.3 (10.5)	Prone
Professional								
Camp et al ¹⁷	Pitchers	42.7 (8.7)	40.9 (7.4)	83.6 (9.7)	42.4 (8.8)	40.8 (8.4)	83.2 (9.8)	Supine
	Nonpitchers	40.6 (7.2)	40.1 (8.7)	80.7 (10.5)	40.1 (9.2)	40.6 (8.6)	80.6 (10.0)	Supine
Camp et al ²²	Pitchers	46.9 (9.8)	39.1 (9.0)	85.9 (9.1)	48.7 (9.9)	37.1 (9.7)	85.8 (10.3)	Supine
	Nonpitchers	45.9 (10.4)	34.1 (11.3)	80.0 (10.4)	46.9 (11.4)	34.0 (10.8)	80.7 (9.3)	Supine
Laudner et al ³⁵	Pitchers	40.9 (8.1)	34.4 (6.1)	75.3 (7.8)	41.0 (6.3)	34.6 (4.4)	75.6 (5.9)	Seated
	Nonpitchers	42.7 (6.8)	37.0 (4.8)	79.7 (7.7)	41.9 (7.3)	37.7 (5.8)	79.6 (7.6)	Seated
Sauers et al ²⁶	Pitchers	30.1 (5.4)	37.0 (5.6)	67.1 (8.3)	30.9 (5.9)	37.2 (5.7)	68.1 (7.7)	Seated
	Nonpitchers	31.5 (4.5)	38.4 (4.7)	70.4 (6.5)	33.8 (4.5)	36.9 (4.9)	70.8 (6.8)	Seated
Robb et al ⁴¹	Pitchers	35.6 (5.8)	31.3 (6.2)	67.0 (7.7)	44.0 (9.0)	50.8 (9.2)	94.8 (13.3)	Prone

Abbreviations: ER, external rotation; IR, internal rotation; NR, not reported; TROM, total rotational range of motion.

an effect of limb dominance on hip extension strength, with the stance limb demonstrating greater hip extension strength in both studies. However, these differences were not significant between limbs. Conversely, Shimamura et al found no differences between stance (90.9 [24.6] kg) and lead (90.6 [20.3] kg) hip extension. One study examined isometric hip flexion in college baseball pitchers and found no significant differences between stance (530.2 [82.3] N) and lead (524.4 [124.0] N) limbs.⁴⁶ One study investigated hip adduction in

college baseball pitchers and found significantly increased strength on the stance limb (392.9 [116.4] N) compared with the lead limb (352.2 [82.3] N).⁴⁶ Finally, a hip asymmetry ratio (stance limb hip abduction strength/lead limb hip abduction strength multiplied by 100) was described in 1 study including professional pitchers and nonpitchers.⁴⁰ The findings in this study demonstrated a hip asymmetry ratio of 10% in pitchers and 10.6% in noninjured pitchers, with 47.7% of pitchers and 48.8% of nonpitchers demonstrating a stronger

Table 5 Hip Strength (ABD, ER, and IR) Across All Levels of Play

Study	Level of play	Position group	Unit of measure	Lead ABD	Stance ABD	Lead ER	Stance ER	Lead IR	Stance IR
Beckett et al ³⁰	Youth	Pitchers and nonpitchers	N/kg	12.6	12.6	NR	NR	NR	NR
Harding et al ³²	Youth	Pitchers	kg	14.1 (3.3)	13.6 (2.2)	8.5 (2.9)	7.9 (2.1)	8.7 (2.8)	9.0 (3.0)
Oliver et al ³⁸	Youth	NR	N	NR	NR	118.6 (50.6)	119.0 (42.7)	132.3 (49.4)	129.0 (56.8)
Beckett et al ³⁰	High school	Pitchers and nonpitchers	N/kg	17.8	17.4	NR	NR	NR	NR
Shimamura et al ⁴⁵	College	Pitchers	kg	40.6 (10.9)	42.9 (10.2)	38.1 (8.5)	36.5 (8.8)	55.3 (13.6)	49.2 (13.2)
Yanagisawa et al ⁴⁶	College	Pitchers	N	452.6 (121.0)	469.2 (116.4)	120.5 (32.3)	129.4 (34.7)	113.1 (30.6)	110.7 (32.0)
Zeppieri et al ⁴⁷	College	Pitchers	N·m	107.0 (10.6)	109.0 (16.4)	NR	NR	NR	NR
Laudner et al ³⁴	Professional	Pitchers	kg	41.9 (7.2)	41.4 (6.3)	NR	NR	NR	NR
	Professional	Nonpitchers	kg	43.0 (7.2)	44.9 (6.6)	NR	NR	NR	NR
Zipser et al ⁴⁹	Professional	Pitchers	N·m	163.3 (39.4)	166.3 (35.0)	NR	NR	NR	NR
	Professional	Nonpitchers	N·m	191.8 (50.4)	196.6 (42.0)	NR	NR	NR	NR

Abbreviations: ABD, abduction; ER, external rotation; IR, internal rotation; NR, not reported.

lead than stance leg. One study included strength measures but did not have extractable strength data and therefore were excluded from the results.²⁵

Discussion

The normative hip ROM and strength data presented in this review can be used to monitor and inform health care and performance professionals in the care of baseball players across multiple levels of play. Clinicians must consider the effects of limb dominance in baseball athletes and in turn, discrepancies between the stance and stride limbs when ascertaining clinical data regarding hip ROM and strength; as such, comparing clinical populations to similar populations of interest (ie, baseball athletes of similar level of play and position) are an important consideration in the evaluation of the overhead athlete. With this in mind, the present authors sought to provide a comprehensive resource for normative hip musculoskeletal profiles in baseball athletes for clinical consideration. Integration of these variables into a screening protocol garners sports medicine professionals the opportunity to monitor for baseball-specific hip ROM and strength imbalances and to compare athletes within similar populations.

Hip ROM

Of the 4 groups of baseball athletes included in this review, articles that included a youth cohort yielded the most results. By position, 4 studies measured hip ER and IR in the supine position,^{36,42–44} 4 in the prone position,^{25,30,32,42} and 1 in the seated position.³⁷ Given the variability in testing position, overall trends in the data were difficult to interpret. However, 1 general theme noted was that, when measured in the prone position, youth baseball athletes tend to demonstrate more hip IR ROM than hip ER ROM on both their lead and stance limbs, regardless of position played. When looking at youth athletes measured in the supine position, there is more discrepancy in the data. Of the 4 studies that utilized the supine

position, 1 study reported greater hip IR on both the stance and lead limbs,³⁶ 2 studies reported greater hip ER on both the stance and lead limbs,^{42,44} and 1 study only measured hip IR.⁴² When measured in the seated position, Oliver et al^{37,38} found both lead and stance limb hip ER to be more than that of hip IR. When looking across testing positions, reported ranges in the supine position were consistently higher for both hip ER and IR, while the prone position demonstrated the lowest values of hip ER, and the seated position demonstrated the lowest values of hip IR. Overall, lead hip ER demonstrated the largest ROM on average of 40.9° regardless of position, while lead hip IR has the lowest average of 37.8°. It is important to note discrepancies in methodology beyond testing position. Seven total studies utilized a standard or bubble level goniometer,^{25,30,32,36,42–44} while the remaining 2 used a digital inclinometer.^{37,38} Only 5 studies including youth athletes reported reliability measures for the rater obtaining hip ROM data.^{25,32,37,38,42}

In the high school baseball population, 1 study measured hip ER and IR ROM in the supine position,³¹ 2 in the prone position,^{25,30} and 1 in the seated position.³⁹ Similar to that of the youth baseball population, individuals measured in supine and seated demonstrated more lead and stance hip ER than IR. However, the 2 studies that measured hip ROM in the prone position have conflicting evidence.^{25,30} Beckett et al³⁰ reported an increase in stance and lead limb hip ER, while Picha et al²⁵ reported increased lead and stance limb hip IR. Similar to the youth population, high school athletes who were measured in seated position demonstrate increased hip ER as compared with hip IR for both the stance and lead limbs. When looking across testing positions, athletes tested in the supine position had the greatest total rotational arc of motion compared with the prone and seated testing positions. Overall, the identified evidence including a cohort of high school baseball players is limited, garnering it difficult to suggest definitive pooled trends. Of the 4 articles included that reviewed the high school baseball population, 3 studies used a standard or bubble level goniometer to collect hip ROM data,^{25,30,31} and only 2 studies reported rater reliability.^{25,39}

A total of 8 studies included a college baseball cohort, 2 studies measured hip ER and IR in supine,^{24,33} 6 in prone,^{33,35,45–48} and 2 in

seated.^{23,45} Similar to that of the youth and high school baseball populations, individuals measured in the seated position demonstrated increased lead and stance limb hip ER when compared with IR. However, when measuring hip ROM in the supine position, hip rotational ROM values differed from those measured in the seated position. When measured in the supine position, lead and stance limb hip ER and IR were nearly symmetric from side to side. These results of near-symmetric hip ER and IR on both the stance and lead limbs in a college cohort differed from all other cohorts measuring hip ROM in the supine position. The hip ROM values published from the prone testing position differed from that of a youth or high school population. In a college cohort, baseball athletes were observed to have increased stance and lead limb hip ER when compared with hip IR, demonstrating increased lead and stance limb ER compared with IR. These cumulative findings are of interest and potentially demonstrate the need to consider how osseous development and maturation may affect hip ROM changes across competitive levels. In this population, 4 studies used a standard or bubble level goniometer for data collection,^{24,46–48} and 6 studies reported rater reliability.^{23,33,35,45–47}

Last, in the professional baseball cohort, 2 studies measured hip ER and IR in the supine position,^{17,22} 1 in the prone position,⁴¹ and 2 in the seated position.^{26,34} Similar to that of the youth, high school, and college baseball populations, individuals measured in the supine and seated positions demonstrated increased lead and stance limb hip ER than hip IR, with the exception of one study, whose results reported the opposite.²⁶ In the single study that measured in the prone position, pitchers demonstrated more lead hip ER than IR but increased stance limb hip IR ROM than ER.⁴¹ Robb et al⁴¹ reported stance limb hip IR at 50.8° on average, which is 10° more than any other reported study in this patient population. If the findings from this aforementioned study were excluded, a similar theme continues across all included populations: Baseball athletes typically present with more hip ER ROM when compared with hip IR, regardless of laterality. In this population, 4 studies utilized a goniometer to collect their data,^{17,22,26,41} only one studies reported rater reliability.⁴¹

In looking at the results across all levels of play, we consistently see that individuals demonstrate more ER on both their lead and stance limbs compared with IR. This theme holds true except when looking at professional baseball players; however, this may be skewed by the findings from the study by Robb et al.⁴¹ In general, the present authors note a trend across increasing levels of play and, in turn, age. We observe larger differences between hip ER than IR ROM as individuals mature and transition from youth through college. When pooling reported means for hip rotational ROM across all studies, baseball athletes demonstrate approximately 40° of hip ER on both the stance and lead limbs and 35° of hip IR on both the stance and lead limb. It is important to note that these pooled means are simply observations, and no formal analysis was conducted to determine differences between levels of play nor by measurement position. There are vast inconsistencies in the standardization techniques for measuring and reporting ROM values for both measuring position and instrumentation selected. The most common position measured in all of the studies was the prone position (16 total studies), with the supine position being the second most common (11 total studies), and the seated position being the least utilized (9 total studies). The most used instrumentation was the goniometer (standard or bubble level), followed by a standard or digital inclinometer.

Isometric Hip Strength

Of the 26 articles included in this review, less than half reported assessing any form of lower-extremity strength in baseball

players.^{25,30,32,34,38,40,45–48} Hip strength data were available in 10 of the research studies, with hip abduction measures being most often recorded.^{25,30,32,34,38,40,45–47,49} Hip IR and ER strength measures were also assessed by multiple authors,^{25,32,38,45,46} while hip flexion,⁴⁶ adduction,⁴⁶ and hip symmetry index values⁴⁰ were examined in single studies only. All strength measures were assessed isometrically using a handheld dynamometer; however, positions and study protocols varied. The paucity of evidence and lack of consistency among study methods and outcomes of interest made it difficult to draw any valid conclusions on how best to assess and interpret hip strength values to prevent injuries and maximize performance in baseball populations.

Previous research examining hip strength in athletic populations have shown that factors, such as age and physical size (ie, height and weight) directly impact the forces recorded by a muscle or muscle group.^{50–52} Beckett et al³⁰ and Picha et al²⁵ examined hip strength measures in 2 cohorts of youth and adolescent baseball players. Both studies found that the older, more physically mature players demonstrated greater hip strength values when compared with their younger counterparts.^{25,30} In an effort to control for the effects of physical size on force production, Beckett et al³⁰ normalized hip strength data using player bodyweight, while Picha et al²⁵ did not.

Sport participation factors have also been correlated with hip strength measures, particularly in youth pitchers.³² Harding et al³² found that years of competitive play was positively correlated with increased hip IR strength in both the stance and lead legs as well as increased hip abduction strength in the lead leg. Number of months played in the last year was also associated with increased hip IR strength in the stance leg. Hip strength values were not normalized to bodyweight or limb length in this study.³² In the college and professional baseball literature, differences in hip strength values have been assessed based on player position and arm dominance.^{34,45,49} Laudner et al³⁴ compared hip strength in pitchers and nonpitchers and found that pitchers demonstrated less hip abduction strength in the stance leg when compared with position players; however, players' physical size was not taken into account. Another study noted no differences in stance or lead leg hip strength between left hand dominant and right hand dominant college pitchers; however, these data were unavailable at the youth and professional levels.⁴⁵

Apart from the limited physical and sport participation characteristics reported in conjunction with hip strength assessment in baseball players, significant variations in testing procedures were also noted. Hip abduction strength was consistently assessed in a side-lying position across all 10 studies; however, knee position did vary ultimately, impacting dynamometer placement and force production (ie, flexed = distal thigh and extended = lateral malleolus).^{25,30,32,34,38,40,45–47,49} Hip IR and ER strength was measured either seated or prone with the knee flexed to 90° in both positions allowing consistent dynamometer placement.^{25,32,38,45,46} However, the variations in hip angle between the seated and prone testing positions significantly limited any comparisons within the data reported. While make-test methods were most often used to assess isometric hip strength in baseball players,^{32,38,40,45–47,49} 3 studies did use the less reliable break-test method.^{25,30,34} Emphasizing the importance of consistency and reliability of testing methods moving forward will significantly improve clinicians' ability to utilize hip strength data as a part of injury prevention and performance programs.

Throwing-related pain and/or injury and ball velocity are 2 of the main outcomes of interest in baseball research. Shoulder ROM

and strength measures have been well documented in relation to baseball outcomes^{53–55}; however, few studies exist examining the impact of hip strength.^{38,40,46} One group of authors found a positive correlation between ball velocity and stance and stride leg hip strength in a cohort of healthy college pitchers.⁴⁶ The pitchers who threw harder off the mound demonstrated greater hip strength than those with lower ball velocities. Two studies have examined the relationships between hip strength measures and throwing arm pain/injury in baseball players.^{38,40} Plummer et al⁴⁰ found that professional pitchers who demonstrated asymmetries in stance and lead leg hip strength were more likely to sustain a shoulder or elbow injury when compared with pitchers who did not. This relationship was not observed in youth baseball populations.³⁸ Increased standardization in hip strength assessments is imperative for baseball research moving forward. Efforts to normalize data based on physical size (ie, bodyweight and limb length) should be encouraged, thus allowing clinicians to better understand how hip strength develops and responds to sport-related stress over time. More robust studies are needed in baseball to determine how factors, such as age, participation level, and player position impact hip strength measures and ultimately injury prevention and athletic performance.

Limitations

Research synthesis of hip ROM and strength variables of baseball athletes is difficult given the diversity of players across the competitive spectrum (age, skill level, injury history, and playing volume). It is unclear how injury history, baseball-specific playing volume, or other sports activity may contribute to the variables of interest present in this review. Several inconsistencies were noted across the included studies, namely in the methodology, including rater experience/reliability, instrumentation, testing position, and reported units for outcome variables. These inconsistencies are a major limitation in collectively pooling and analyzing the available data, even qualitatively. While the authors of this review made a conscious effort to diminish the risk of bias through dual author appraisal, only one author ran the initial searches for article inclusion. Additionally, the data extracted were not compared between each of the authors; all data were reviewed rigorously on multiple occasions to ensure accuracy. The current data available on the normative hip ROM and strength profiles in baseball athletes emphasizes youth and professional athletes more so than college and high school athletes. Last, baseball has evolved over time in terms of training philosophies, volume of gameplay, and position-specific specialization; readers must take this into consideration when comparing earlier publications (2010) with later (2024) publications.

Conclusion

The unique demands of the entire kinetic chain during overhead throwing likely results in hip ROM and strength changes between limbs, between positions, and across levels of play. Overall, the data extracted for these purposes demonstrates that hip ROM and strength values differ across level of play, between limbs, and between positions. These data may be useful for screening purposes for baseball athletes and may aide in the plan of care development in the meaningful intervention to improve observed deficits or imbalances in hip ROM or strength. Clinicians should strongly consider their personal clinical methodology and populations of interest prior to comparing clinical data to the data

presented in this review to ensure similar methodology and patient populations are appropriately referenced as “normative” data. Additionally, longitudinal studies are needed to make evidence-based recommendations for performance improvement or injury risk mitigation. Last, sports medicine professionals and researchers are urged to use previously established methods consistently across studies to garner clinicians the opportunity to make more direct comparisons between populations.

Acknowledgments

Protocol Registration: Protocol registration was denied by PROSPERO due to the title and content being unrelated to COVID-19 studies. Further explanation and documentation of correspondence with JSR editorial staff can be directed to the corresponding author of this manuscript.

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